

Axiom Switch Universe Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Extended	ID	axiom-switch-universe

Big idea: Axioms create mathematical worlds.

Overview

Students explore how changing assumptions changes mathematical truth.

Student Prompt

Inspect Axiom Switch Universe Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Axiom Switch Universe Stimulus A	Student-facing stimulus 1: A geometry rule change where parallel lines behave differently. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Axiom Switch Universe Stimulus B	Student-facing stimulus 2: A made-up arithmetic system with a changed operation rule. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Axiom Switch Universe Reveal / Twist	Student-facing stimulus 3: A reflection on whether certainty depends on starting assumptions. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Axioms create mathematical worlds.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Axiom Switch Universe Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Axiom Switch Universe Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Are mathematical truths absolute?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how axioms create mathematical worlds.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A geometry rule change where parallel lines behave differently.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A geometry rule change where parallel lines behave differently.
- A made-up arithmetic system with a changed operation rule.
- A reflection on whether certainty depends on starting assumptions.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups different geometry worlds: exactly one parallel line, no parallel lines, or many parallel lines through a point not on a line.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students draw what triangles, lines and maps might be like in their world.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Each group states one theorem that may change.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Discuss usefulness, consistency and truth.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Proof versus Pattern
- Monty Hall Shock
- Calculator Trust Test